

mistake, also raising $U_j^{(2)}$. Strategic complementarities capture an important dimension of social media interactions — utility feedback from others' behavior tends to encourage agents to cohere with those behaviors.

Equilibrium Structure. The next theorem shows that an equilibrium always exists and is in cutoff strategies. At the same time, strategic complementarities ensure that there is a well-defined structure to the set of equilibria. To make this more concrete, we say that an equilibrium $(\mathbf{b}^*, \mathbf{b}^{**})$ has *uniformly more sharing* than other $(\hat{\mathbf{b}}^*, \hat{\mathbf{b}}^{**})$ if $\mathbf{b}^* \preceq \hat{\mathbf{b}}^*$ and $\mathbf{b}^{**} \preceq \hat{\mathbf{b}}^{**}$ (where $\preceq$ is the component-wise order). This, in particular, means that all the thresholds for each agent is (weakly) lower in the former equilibrium (recall that lower thresholds mean more sharing).

Theorem 1.

- (i) *There exists a Bayesian-Nash equilibrium;*
- (ii) *All equilibria are in cutoff strategies;*
- (iii) *The set of cutoffs $(\mathbf{b}^*, \mathbf{b}^{**})$ forms a lattice, and thus there exists a least-sharing and most-sharing equilibrium.*

The structure of equilibria characterized in Theorem 1 facilitates our analysis, enabling us to focus on two sets of thresholds, $(\mathbf{b}^*, \mathbf{b}^{**})$, which are themselves monotonic in the reliability of the article in question. Because of strategic complementarities, there can be multiple equilibria: when others are choosing to share an article with middling reliability, this further encourages sharing because one's own post will circulate more, increasing the utility from sharing. Conversely, if the same article with middling reliability is not shared by others, the payoff to sharing is reduced, while the cost of being found out to have circulated misinformation remains constant. This then discourages sharing.

Theorem 1 also shows that, despite this multiplicity, there are two focal equilibria on which we can concentrate: the equilibrium with the smallest vector of cutoffs (*most-sharing equilibrium*) and the equilibrium with the largest vector of cutoffs (*least-sharing equilibrium*).

Finally, the theorem's characterization provides an explicit measure of the amount of sharing. Recall that agent i 's prior belief b_i is drawn *ex ante* from the distribution H_i . Hence, in an equilibrium with cutoff b_i^{**} for agent i , the *ex ante* likelihood that this agent will share is $1 - H_i(b_i^{**})$. Thus, clearly, the most sharing equilibrium, which has the smallest equilibrium b_i^{**} for all i , has the highest likelihood that any agent i will share the article in question.

3.2 Comparative Statics (for Fixed P)

In this subsection we provide comparative statics for the most and least sharing equilibria as we change the parameters of the social media game, but holding the sharing network $\mathbf{P}$ fixed. We discuss comparative statics with respect to the network in the next section. Throughout this subsection, we say that there is *uniformly more sharing* if sharing increases in both the most sharing and the least sharing equilibria.